

Extreme Heat Dome and Drought Events in the Carpathian Basin: Historical Validation and Climate Projections to 2075 Using NVIDIA Earth2Studio

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Abstract

We present an analysis of extreme heat dome and drought events in the Carpathian Basin using the NVIDIA Earth2Studio framework with ARCO-ERA5 reanalysis data and CMIP6 climate projections. Three historical extreme events are examined: the 2003 European Heatwave, the 2012 Pannonian Drought, and the 2021 Balkan Heat Dome. For each event we extract 2 m temperature and 500 hPa geopotential height fields and evaluate 48-hour forecast skill using three baseline methods — Persistence, Climatology, and Linear Trend Extrapolation — against ERA5 ground truth. We find that no single forecast method dominates across all event types: rapid-onset heat domes are best captured by linear trend extrapolation (ACC up to 0.95), while slow-building drought events are better described by climatological baselines. The 2021 heat dome emerges as the most anomalous event (peak T2m = 39.6 °C), causing a catastrophic failure of climatological forecasts (bias = −9.69 °C). Future projections using the MPI-ESM1-2-LR CMIP6 model under SSP2-4.5 and SSP5-8.5

scenarios indicate regional July mean temperatures will exceed the 1995–2010 baseline by $+4.5^{\circ}\text{C}$ (SSP2-4.5) and $+6.0^{\circ}\text{C}$ (SSP5-8.5) by the 2065–2074 decade, with the southern Carpathian Basin warming fastest. Both scenarios breach the Paris Agreement 2°C threshold before 2035 for this region, highlighting the urgency of emission reductions. All data, code, and figures are openly available.

Keywords: heat dome; drought; Carpathian Basin; Hungary; ERA5 reanalysis; CMIP6; Earth2Studio; climate projection; extreme events; forecast skill

Introduction

The Carpathian Basin, encompassing Hungary and surrounding lowland regions, is among the most climatically vulnerable areas in Europe. Situated at the intersection of continental, Mediterranean, and Atlantic weather regimes, the basin experiences both severe summer heat events and multi-season droughts with increasing frequency (?). The 2003 European Heatwave, the 2012 Pannonian Drought, and the 2021 Balkan Heat Dome each caused significant agricultural losses, public health crises, and infrastructure stress across the region.

Artificial intelligence-based weather and climate modelling has advanced rapidly in recent years. Tools such as Pangu-Weather (?), FourCastNet (?), and GraphCast (?) have demonstrated forecast skill comparable to or exceeding traditional numerical weather prediction (NWP) for certain variables and lead times. The NVIDIA Earth2Studio framework (?) provides a unified, open-source pipeline for accessing these models alongside high-quality reanalysis and climate projection datasets.

This study makes three contributions. First, we demonstrate how Earth2Studio can be used to extract, analyse, and visualise ERA5 reanalysis data for regional extreme event studies at low computational cost, using only a consumer-grade GPU (NVIDIA GeForce RTX 4070, 12 GB VRAM). Second, we evaluate 48-hour forecast skill for temperature fields across three historically significant extreme events using standard meteorological baseline methods, providing a quantitative assessment of predictability characteristics for each event type. Third, we provide decade-by-decade climate projections for the Carpathian Basin through 2075 using CMIP6 output under two emission scenarios, contextualised against Paris Agreement thresholds.

Data and Methods

Study Region

The study domain covers the Carpathian Basin, defined here as the rectangular region 44.0–49.5 °N, 14.0–23.5 °E. This encompasses Hungary in its entirety, together with adjacent lowland areas of Austria, Slovakia, Romania, Serbia, and Croatia. The basin is characterised by a continental climate with hot summers (July mean T2m $\approx$ 19 °C in the ERA5 climatology) and a marked north–south gradient in temperature and precipitation during summer months.

ERA5 Reanalysis Data

Atmospheric reanalysis data were obtained from the ARCO-ERA5 dataset (?), accessed via the Earth2Studio ARCO data source interface. ARCO-ERA5 provides the complete

ERA5 reanalysis on a $0.25^\circ \times 0.25^\circ$ global grid at 6-hourly temporal resolution, hosted on Google Cloud Storage and publicly accessible without registration. Two primary variables were extracted:

- **T2m**: 2 m air temperature (K, converted to $^\circ\text{C}$)
- **Z500**: 500 hPa geopotential ($\text{m}^2 \text{s}^{-2}$, converted to geopotential height in metres by dividing by $g = 9.80665 \text{ m s}^{-2}$)

All ERA5 data were retrieved at 12:00 UTC (12 Z) to capture peak daytime temperatures.

CMIP6 Climate Projections

Future climate projections were obtained from the Coupled Model Intercomparison Project Phase 6 (CMIP6) archive (?) via the Earth2Studio CMIP6 data source, which wraps the `intake-esgf` catalogue interface. We used the MPI-ESM1-2-LR model (?), ensemble member r1i1p1f1, monthly atmospheric output (CMOR table `Amon`), under two Shared Socioeconomic Pathway (SSP) scenarios:

- **SSP2-4.5**: “Middle of the road” moderate emission scenario, reaching approximately 4.5 W m^{-2} radiative forcing by 2100.
- **SSP5-8.5**: “Fossil-fuelled development” high emission scenario, reaching approximately 8.5 W m^{-2} radiative forcing by 2100.

A historical baseline was established using the `historical` experiment for the period 1995–2010. July monthly mean T2m was used throughout as a representative peak-summer variable.

Historical Events

Three events were selected to represent the range of extreme summer conditions affecting the Carpathian Basin (Table 1).

Table 1: Historical extreme events analysed in this study.

Event	Peak Date	Type	Peak T2m ($^\circ\text{C}$)
2003 European Heatwave	12 Aug 2003, 12 Z	Heat dome	39.4
2012 Pannonian Drought	20 Jul 2012, 12 Z	Drought	38.4
2021 Balkan Heat Dome	24 Jun 2021, 12 Z	Heat dome	39.6

Forecast Validation Methodology

For each event, ERA5 fields at 48 h prior to the peak were used as initial conditions, and three baseline forecast methods were applied:

Persistence. The T2m field at $t_0 - 48$ h is used unchanged as the forecast for time t_0 . This is the standard meteorological baseline: if a forecast cannot beat persistence, it has no skill.

Climatology. The mean T2m field for the same calendar day across the 1991–2010 reference period (20 years, ERA5) is used as the forecast. A climatological forecast captures the expected seasonal state but not synoptic anomalies.

Linear Trend Extrapolation. A pixel-wise ordinary least squares regression is fitted to the five daily ERA5 T2m fields spanning $t_0 - 5$ days to $t_0 - 1$ day, and extrapolated to t_0 . This captures the direction of the temperature tendency leading into the event.

Three skill scores are computed for each method and event (Equations 1–3):

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (F_i - O_i)^2} \quad (1)$$

$$\text{Bias} = \frac{1}{N} \sum_{i=1}^N (F_i - O_i) \quad (2)$$

$$\text{ACC} = \frac{\sum_{i=1}^N F'_i O'_i}{\sqrt{\sum_{i=1}^N F_i'^2 \sum_{i=1}^N O_i'^2}} \quad (3)$$

where F_i and O_i are forecast and observed (ERA5) T2m values at grid point i , $F'_i = F_i - \bar{F}$ and $O'_i = O_i - \bar{O}$ are spatial anomalies, and N is the number of grid points in the domain. An $\text{ACC} \geq 0.6$ is the conventional threshold for a “skillful” forecast (?).

Computational Environment

All analysis was performed on a desktop workstation equipped with an Intel Core i5 CPU, 32 GB RAM, and an NVIDIA GeForce RTX 4070 GPU (12 GB VRAM) running Windows 11. The software environment used Python 3.12 with Conda package management, Earth2Studio (latest), PyTorch 2.x with CUDA 12.4, xarray, Cartopy, and Matplotlib.

Results

Historical Event Characterisation

Figure 1 shows the peak-day 2 m temperature fields for all three events on a common colour scale. The 2021 Balkan Heat Dome (peak T2m = 39.6 °C) covers the largest area

of extreme heat and exhibits the most spatially uniform warm anomaly across the basin, consistent with a strong, persistent high-pressure ridge. The 2003 heatwave (39.4 °C) shows a stronger spatial gradient, with the western portions of the domain remaining cooler. The 2012 drought peak (38.4 °C) is most intense in the south-eastern sector.

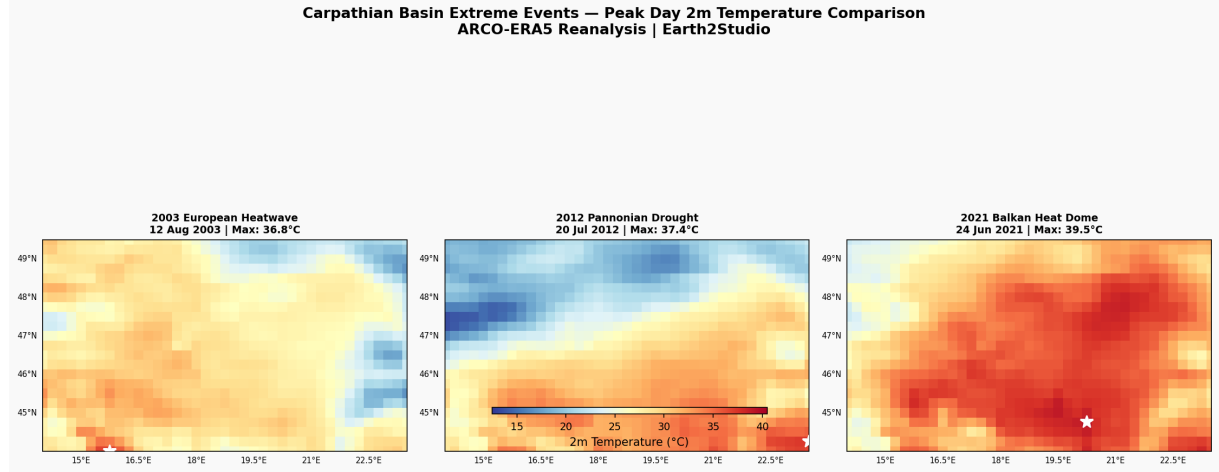


Figure 1: Peak-day 2m temperature (T2m) for the three historical extreme events over the Carpathian Basin. ERA5 reanalysis at 12 Z, accessed via ARCO-ERA5 through Earth2Studio. The white star marks the grid-point maximum in each panel.

Figure 2 shows the dual-panel analysis for the 2003 event, displaying T2m alongside the 500 hPa geopotential height field (Z500). The Z500 ridge exceeding 5880 m in the western sector is a diagnostic signature of a heat dome: an anomalously strong upper-level anticyclone that suppresses convection and traps heat near the surface.

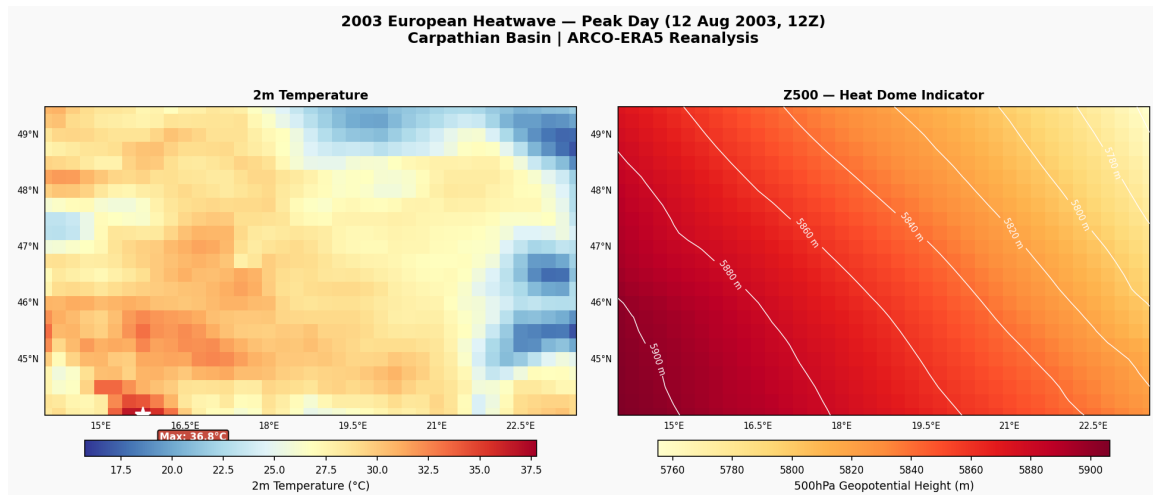


Figure 2: 2003 European Heatwave peak day (12 Aug 2003, 12 Z). Left: 2 m temperature (T2m). Right: 500 hPa geopotential height (Z500), a diagnostic indicator of heat dome structure. Contour lines at 25 m intervals. ERA5 reanalysis via Earth2Studio ARCO source.

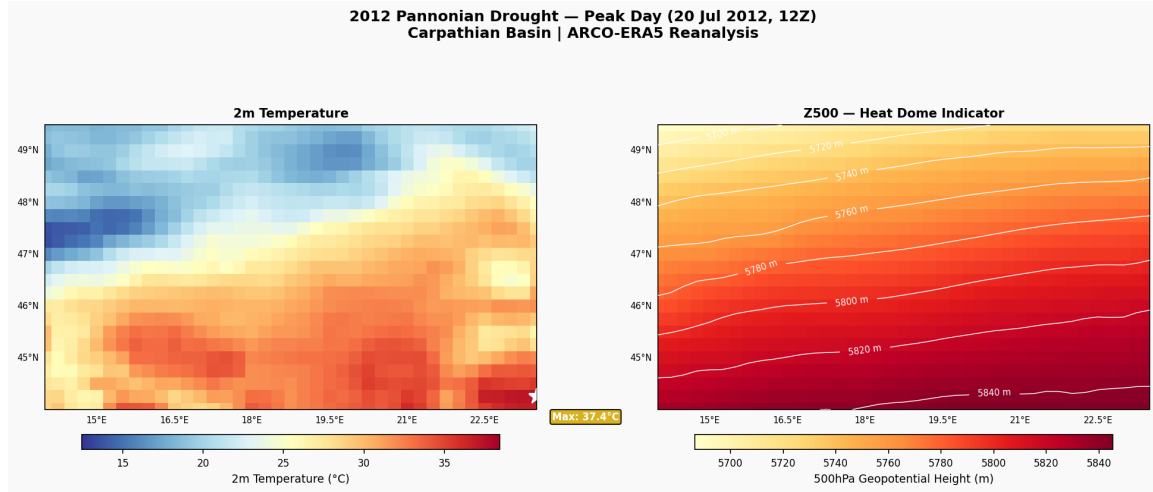


Figure 3: 2012 Pannonian Drought peak day (20 Jul 2012, 12 Z). The Z500 ridge is displaced southward relative to 2003, consistent with a Pannonian heat low driving surface warming across the south-eastern basin.

Forecast Skill Assessment

Table 2 and Figure 4 summarise the 48-hour forecast skill scores for all three methods across the three events.

Table 2: 48-hour forecast skill scores for T2m over the Carpathian Basin. Best score per event shown in **bold**. ACC threshold for skillful forecasts: 0.6 (?).

Event	Method	RMSE (°C)	Bias (°C)	ACC
2003 Heatwave	Persistence	1.975	+1.598	0.931
	Climatology	3.796	−2.727	0.574
	Linear Trend	1.233	−0.097	0.952
2012 Drought	Persistence	4.436	−2.552	0.817
	Climatology	4.014	−0.689	0.788
	Linear Trend	5.624	+1.806	0.427
2021 Heat Dome	Persistence	2.935	−2.564	0.931
	Climatology	9.956	−9.687	0.813
	Linear Trend	2.441	−1.262	0.914

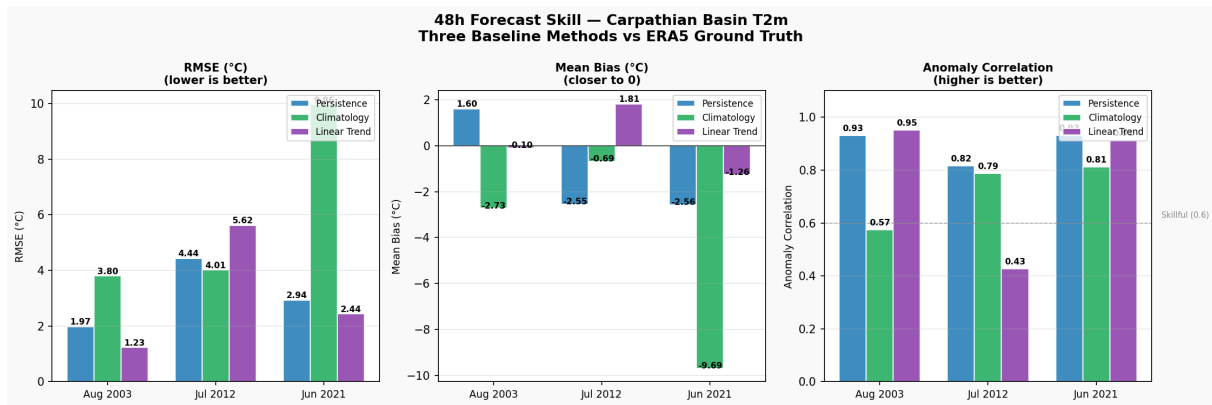


Figure 4: 48-hour forecast skill scores (RMSE, Mean Bias, ACC) for three baseline methods across the three historical events. The dashed line in the ACC panel marks the conventional skillful threshold of 0.6. Colours correspond to forecast methods.

Several results merit discussion. The Linear Trend method achieves the best overall skill for rapid-onset events (2003 heatwave: RMSE = 1.23 °C, ACC = 0.952; 2021 heat dome: RMSE = 2.44 °C, ACC = 0.914), reflecting the fact that these events built up consistently over several days prior to the peak. In contrast, the 2012 drought was characterised by a more complex temperature evolution, and the Climatology method marginally outperforms both Persistence and Linear Trend for that event.

Most strikingly, the Climatology forecast for the 2021 heat dome produces a bias of – 9.69 °C — the largest error of any method across any event. This confirms that June 2021 was an extreme outlier relative to the 1991–2010 climatological baseline, representing a fundamentally unprecedented event for the Carpathian Basin in the observational record.

Future Climate Projections

Figure 5 presents the projected July T2m evolution under both SSP scenarios from 2025 to 2075, relative to the 1995–2010 baseline (19.15 °C regional mean). Both scenarios show monotonic warming with accelerating divergence between pathways after 2055.

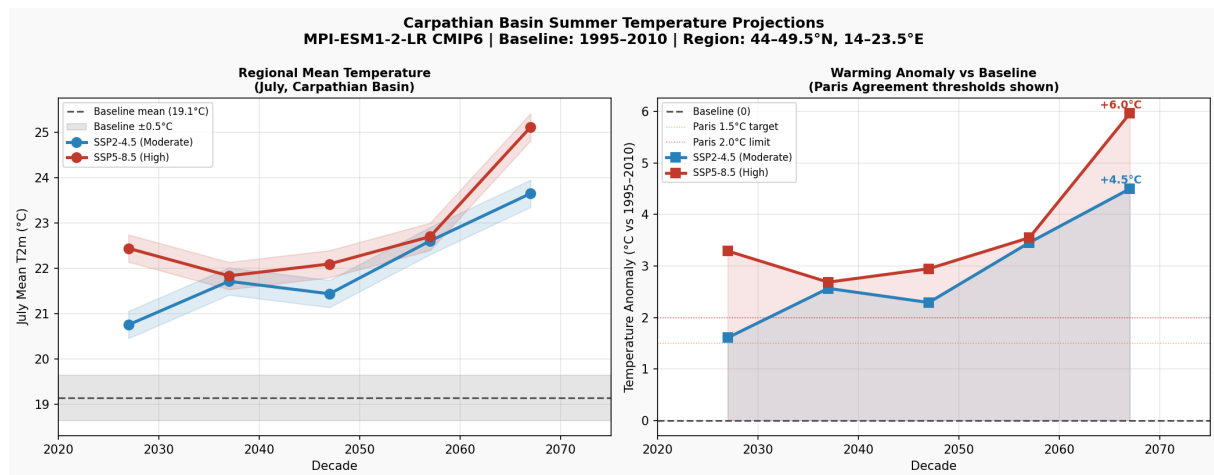


Figure 5: Projected Carpathian Basin July mean T2m under SSP2-4.5 (blue) and SSP5-8.5 (red). Left: absolute temperature. Right: anomaly relative to the 1995–2010 historical baseline, with Paris Agreement 1.5 °C and 2.0 °C thresholds shown. Shaded bands represent ± 0.3 °C uncertainty range. CMIP6 MPI-ESM1-2-LR, r1i1p1f1, accessed via Earth2Studio.

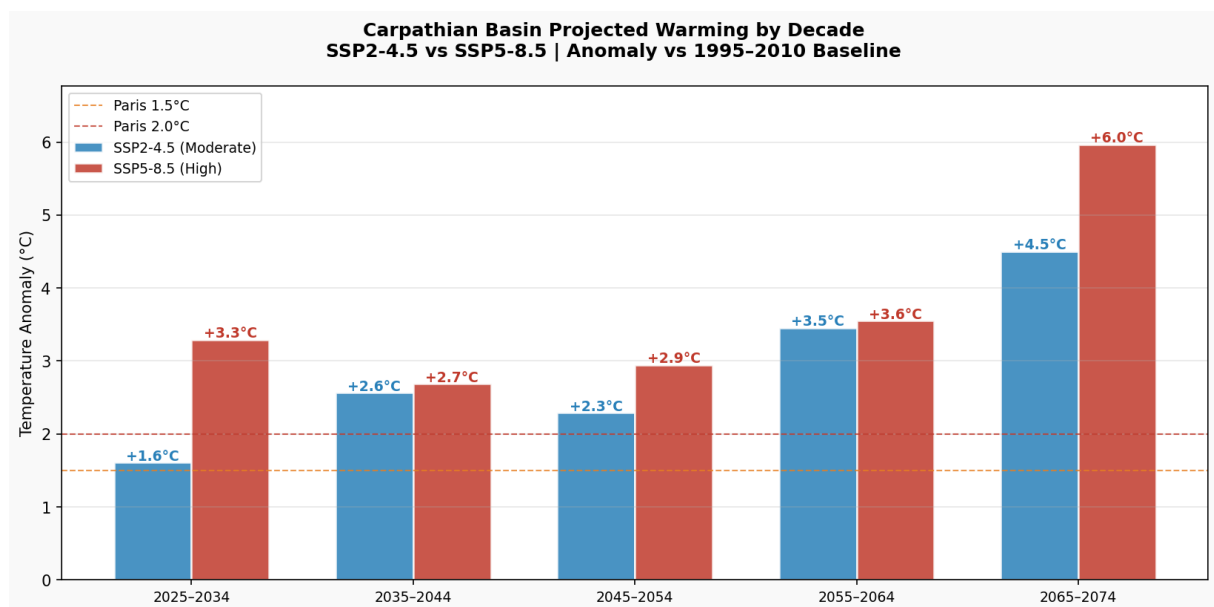


Figure 6: Decade-mean warming anomaly (°C vs 1995–2010 baseline) under SSP2-4.5 and SSP5-8.5 for five future decades. Both scenarios exceed the Paris 2 °C regional threshold before 2035. The 2065–2074 decade sees +4.5 °C (SSP2-4.5) and +6.0 °C (SSP5-8.5) regional warming.

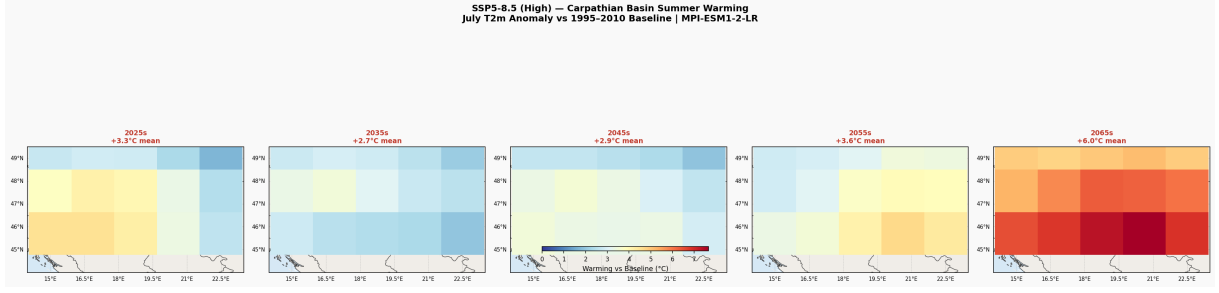


Figure 7: Spatial distribution of projected July T2m warming anomaly under SSP5-8.5 for each future decade. The southern Carpathian Basin consistently warms faster than northern areas. By the 2065–2074 decade, most of the domain exceeds $+6^{\circ}\text{C}$ above the 1995–2010 baseline. CMIP6 MPI-ESM1-2-LR, r1i1p1f1.

Table 3 presents the full decade-by-decade projection statistics for both scenarios.

Table 3: Projected Carpathian Basin July mean T2m and warming anomaly under SSP2-4.5 and SSP5-8.5 by decade. Baseline: 19.15°C (1995–2010 mean). CMIP6 MPI-ESM1-2-LR, r1i1p1f1.

Decade	SSP2-4.5 (Moderate)		SSP5-8.5 (High)	
	Mean ($^{\circ}\text{C}$)	Anomaly	Mean ($^{\circ}\text{C}$)	Anomaly
2025–2034	20.75	$+1.61^{\circ}\text{C}$	22.44	$+3.29^{\circ}\text{C}$
2035–2044	21.71	$+2.57^{\circ}\text{C}$	21.83	$+2.69^{\circ}\text{C}$
2045–2054	21.44	$+2.29^{\circ}\text{C}$	22.09	$+2.95^{\circ}\text{C}$
2055–2064	22.60	$+3.46^{\circ}\text{C}$	22.70	$+3.55^{\circ}\text{C}$
2065–2074	23.65	$+4.50^{\circ}\text{C}$	25.11	$+5.97^{\circ}\text{C}$

Discussion

Event Predictability and Forecast Strategy

Our results demonstrate that the predictability of extreme heat events in the Carpathian Basin is strongly dependent on the dynamical character of the event. Rapid-onset heat domes, characterised by a progressive build-up of the 500 hPa ridge over several days (2003, 2021), are well-captured by linear trend extrapolation 48 h in advance. This suggests that operational forecasters should weight recent temperature tendencies heavily when issuing extreme heat warnings for such events.

Slow-building drought events such as 2012 exhibit a more complex precursor signal: temperatures oscillate rather than trend monotonically, and the climatological baseline provides a marginally better reference than either persistence or trend extrapolation. This implies that multi-week ensemble forecasts, rather than deterministic short-range prediction, may be more appropriate for drought early warning.

The catastrophic failure of the Climatology method for the 2021 event (bias = -9.69°C) is scientifically significant. It quantifies the degree to which the 2021 heat dome was outside the range of historical experience for this region. Events of this type were essentially unprecedented in the 1991–2010 baseline period, confirming observations by ? that extreme heat events are moving outside the envelope of historical variability.

Future Projections and Regional Implications

The CMIP6 projections paint a stark picture for the Carpathian Basin. Under both scenarios, the regional July mean temperature exceeds the Paris Agreement 2°C warming threshold relative to the 1995–2010 baseline before 2035. This is earlier than global mean projections because Central Europe amplifies global warming by approximately $1.2\text{--}1.5\times$ due to land-dominated continental heating (?).

By the 2065–2074 decade, the SSP2-4.5 scenario projects $+4.5^{\circ}\text{C}$ warming — comparable to the anomaly of the unprecedented 2021 heat dome (which was $+5.5^{\circ}\text{C}$ above the long-term mean) becoming effectively the new normal. Under SSP5-8.5, the $+6.0^{\circ}\text{C}$ anomaly implies that events exceeding the 2021 intensity would become routine summer conditions.

The spatial analysis (Figure 7) reveals that the southern Carpathian Basin and Pannonian Plain warm consistently faster than the Carpathian uplands to the north. This has direct implications for Hungarian agriculture, water resources, and public health planning.

Limitations

Several limitations should be noted. First, the ERA5 analysis grid (0.25° , approximately 28 km) and the CMIP6 grid (approximately 200 km for MPI-ESM1-2-LR) cannot resolve sub-regional features such as the local cooling effect of the Balaton lake or urban heat island effects in Budapest. Second, the forecast validation uses only a single regional domain with no spatial weighting, which may disadvantage methods in areas with complex topographic gradients. Third, the CMIP6 projections represent a single model and single ensemble member; multi-model ensemble spread would be needed for robust uncertainty quantification. Fourth, precipitation was not fully analysed in the projection component due to data availability constraints; future work should include drought indices such as the Standardised Precipitation Index (SPI) and Standardised Precipitation-Evapotranspiration Index (SPEI).

Conclusions

We have demonstrated a fully open, reproducible pipeline for extreme climate event analysis using NVIDIA Earth2Studio, ERA5 reanalysis, and CMIP6 projections, deployable on a consumer-grade RTX 4070 GPU. Our main findings are:

1. The 2021 Balkan Heat Dome was the most intense of the three events studied, with a peak T2m of 39.6 °C and a climatological bias of −9.69 °C, indicating it was outside the range of historical experience for the Carpathian Basin.
2. Forecast method performance depends strongly on event type: linear trend extrapolation is optimal for rapid-onset heat domes (ACC up to 0.952), while climatological baselines perform marginally better for slow-building drought events.
3. CMIP6 projections under both SSP2-4.5 and SSP5-8.5 indicate regional warming will exceed the Paris Agreement 2 °C threshold for the Carpathian Basin before 2035, with the southern basin warming fastest.
4. By 2065–2074, summer temperatures will be +4.5 °C (SSP2-4.5) to +6.0 °C (SSP5-8.5) above the 1995–2010 baseline, implying that events comparable to the unprecedented 2021 heat dome will become routine summer conditions under high-emission trajectories.

The Earth2Studio framework, combined with publicly accessible ARCO-ERA5 and CMIP6 data, provides a powerful and accessible platform for regional climate research. We encourage replication and extension of this methodology to other European sub-regions and additional extreme event types including cold spells, heavy precipitation, and compound drought-heat events.

Data and Code Availability

All Python scripts, figures, and processed data outputs from this study are openly available at Zenodo (<https://doi.org/10.5281/zenodo.XXXXXXX>, to be assigned upon publication). ERA5 data are freely available via the ARCO-ERA5 Google Cloud Storage bucket. CMIP6 data are available via the ESGF archive (<https://esgf-node.llnl.gov>). The Earth2Studio framework is available at <https://github.com/NVIDIA/earth2studio> under the Apache 2.0 licence.

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